Semester	S.E. Semester IV – CMPN
Subject	R-Programming
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Experiment Number	01
Experiment Title	Air-Quality Data Cleaning Using R: Management of NA, NULL, and NaN Values
Objectives (Skill Set / Knowledge Tested / Imparted)	• To differentiate between NA, NULL, and NaN values in R • To identify, summarize, and quantify missing data in a realworld dataset • To develop reusable, user-defined functions (missing_summary(), calculate_mode(), clean_variable()) for data cleaning • To apply loops for processing multiple variables without repetitive code • To handle missing numerical data using median imputation and missing categorical data using mode imputation • To implement robust error handling using tryCatch() so the program does not terminate on invalid input • To visualize and export a cleaned dataset for further analysis

Description	This experiment uses the Beijing Multi-Site Air Quality Dataset (Aotizhongxin station) from the UCI Machine Learning Repository, containing 35,064 hourly observations of pollutant (PM2.5, PM10, SO2, NO2, CO, O3) and weather (TEMP, PRES, DEWP, RAIN, wd, WSPM) readings. Due to sensor failures and network interruptions, several fields contain missing values. An R program was developed to import and inspect the dataset with error handling, distinguish between NA,
	NULL, and NaN, generate a missing-value summary, impute missing numerical values using median (via a loop-based reusable approach) and missing categorical values (wind direction) using mode, validate a derived pollution_ratio variable for invalid results (NaN/Infinite), and compare missing-value counts before and after cleaning. The results were visualized using a bar chart and the cleaned dataset was exported as a CSV file.
Program	**GitHub link: https://github.com/Explorer1905/air-quality-rcleaning**
Output	**Variable Missing Before Missing After Values Replaced** PM2.5 925 0 925 PM10 718 0 718 SO2 935 0 935 NO2 1023 0 1023 TEMP 20 0 20 WSPM 14 0 14 wd 81 sing values 0 e cleaning: 81 Total mis efor 7,2 1 → 0 after cleaning. Wind direction mode used for imputation: **NE** Bar chart generated showing missing-value counts before vs. after cleaning (attach screenshot). Cleaned dataset exported as cleaned_air_quality_data.csv.

Conclusion	The R program successfully identified and handled missing values across the Beijing air-quality dataset, correctly distinguishing between NA, NULL, and NaN. Median imputation was applied to numerical variables and mode imputation to the categorical wind-direction variable, reducing total missing values from 7,271 to 0. The clean_variable() function's tryCatch() implementation was verified to gracefully handle invalid inputs — nonexistent columns and categorical variables passed as numeric — without halting execution. This experiment reinforced practical skills in loops, user-defined functions, error handling, and missing-data management, which are essential for realworld data preprocessing prior to statistical analysis or machine learning.